INTERNATIONAL MONETARY FUND

PAPUA NEW GUINEA

REQUEST FOR A STAFF-MONITORED PROGRAM—DEBT SUSTAINABILITY ANALYSIS

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Papua New Guinea: Joint Bank-Fund Debt Sustainability Analysis	
Risk of external debt distress	High
Overall risk of debt distress	High
Granularity in the risk rating	Sustainable
Application of judgement	No

Papua New Guinea (PNG) remains at high risk of debt distress under the Low-Income Country Debt Sustainability Framework (LIC DSF), with weak debt-carrying capacity. While the planned fiscal consolidation helps address debt vulnerabilities exacerbated by the global COVID-19 shock, the risk of both external and public debt distress continues to be assessed as high. Over the medium-term, public debt enters a downward trend and the projected temporary breaches of sustainability indicators can mostly be addressed by debt management operations as well as improvements in revenue generation, a key pillar of the new Staff-Monitored Program (SMP). The Debt Sustainability Analysis (DSA) suggests that PNG is susceptible to exports and exchange rate shocks, signaling serious downside risks to the debt outlook in a global environment of high uncertainty. To lower the risk of debt distress and ensure debt sustainability, gradual fiscal consolidation, including boosting revenues, and steadfast structural reforms to promote private sector growth would be needed. Conditional on the implementation of policy measures included in the SMP, PNG's external and overall debt is judged as sustainable.

[1] The Composite Indicator (CI) of 2.63 is based on the latest available CI information—October 2021 IMF World Economic Outlook (WEO) and the World Bank's Country Policy and Institutional Assessment (CPIA), indicating a "weak" capacity to carry debt.

[2] This Debt Sustainability Analysis has been prepared jointly by the International Monetary Fund and the World Bank, in accordance with the revised Debt Sustainability Framework for low-income countries approved by the Executive Boards of the IMF and the IDA.

PUBLIC SECTOR DEBT COVERAGE

1. The coverage of public debt in the DSA is unchanged from the previous (June 2020) DSA (*Text Table 1*). The segments the public sector captured in the DSA include the central government, state and local government, and to some extent guarantees to other entities in the public and private sector, including parts of state-owned enterprises (SOEs). However, debt numbers do not fully capture implicit government guaranteed debts of SOEs and unfunded superannuation liabilities relating to pensions.[3] For the purpose of this DSA, the coverage of public sector debt remains unchanged from the last DSA, which was prepared in June 2020 in the context of a request to access the IMF's *Rapid Credit Facility* (RCF). Given continued difficulties in capturing and assessing SOE risks, a contingent liabilities stress test is included in this DSA, assuming 9 percent of GDP as SOE debt is not captured in official public debt data, and 3 percent of GDP for other elements of general government (mainly unfunded superannuation liabilities related to pensions. Separately, according to the World Bank's PPP database, the PPP capital stock in PNG is zero and, therefore, no default shock is triggered. A financial market shock of 5 percent is added, reflecting the average fiscal cost of financial crisis in low-income countries. With these assumptions, the cumulative shock in the contingent liabilities stress test amounts to 17 percent of GDP—compared to 7 percent under default assumptions.

Text Table 1. Papua New Guinea: Coverage of Public Debt

	Subsectors of the public sector	Sub-sectors covered
1	Central government	X
2	State and local government	X
3	Other elements in the general government	
4	o/w: Social security fund	
5	o/w: Extra budgetary funds (EBFs)	
6	Guarantees (to other entities in the public and private sector, including to SOEs)	X
7	Central bank (borrowed on behalf of the government)	
8	Non-guaranteed SOE debt	

1	The country's coverage of public debt	The central government plus social security, central bank, government-guaranteed debt		
		Default	Used for the analysis	Reasons for deviations from the default settings
2	Other elements of the general government not captured in 1.	0 percent of GDP	3.0	Unfunded superannuation liabilities, relating to pensions.
3	SoE's debt (guaranteed and not guaranteed by the government) 1/	2 percent of GDP	9.0	SOE sovereign guarantee in dispute.
4	PPP	35 percent of PPP stock	0.0	
5	Financial market (the default value of 5 percent of GDP is the minimum value)	5 percent of GDP	5.0	
	Total (2+3+4+5) (in percent of GDP)		17.0	

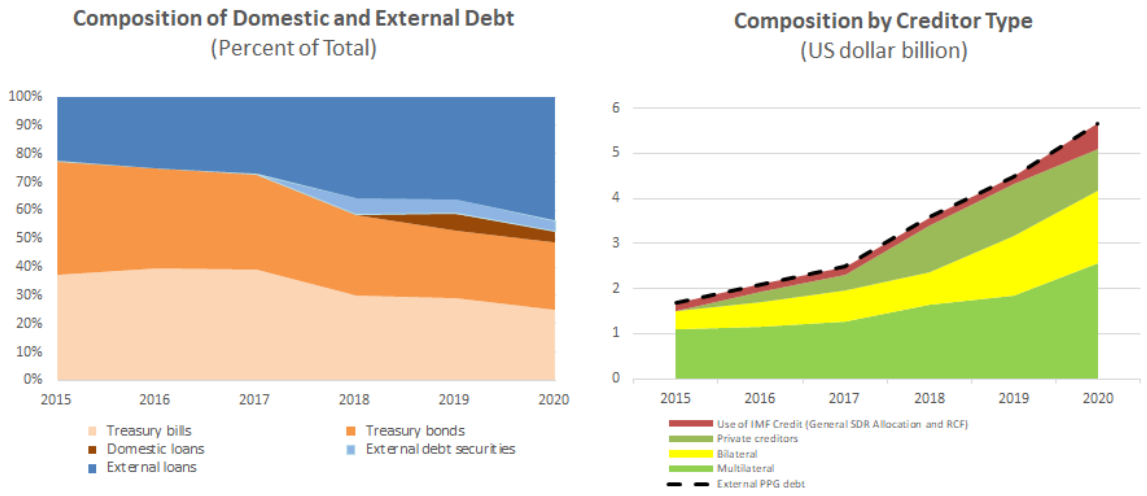
1/ The default shock of 2% of GDP will be triggered for countries whose government-guaranteed debt is not fully captured under the country's public debt definition (1.). If it is already included in the government debt (1.) and risks associated with SoE's debt not guaranteed by the government is assessed to be negligible, a country team may reduce this to 0%.

[3] In the context of the previous SMP, which expired in end-June 2021, the authorities initiated an SOE reform program to reduce the backlog of audited annual financial statements and to strengthen SOE oversight and improve understanding of fiscal risks. This program includes a detailed review of SOE debt and government guarantees and is also expected to improve the reporting of public debt.

BACKGROUND ON DEBT

2. During 2017-2020, public debt build-up was mainly due to external loans and the creditors composition has been shifting from commercial loans to official multilateral and bilateral financing. Domestic financing through T-bills and T-bonds has been stable over time. *Text Figure 1* shows the evolution of domestic and external debt, broken down into instruments (left chart), and the composition of debt by creditor group (right chart) with categories official multilateral, official bilateral Paris Club, official bilateral Non-Paris Club, commercial lenders, and use of IMF credit. Shocks from the pandemic, including terms-of-trade shocks to main exports (mainly LNG and metals), have reduced the 2020 current account surplus and exacerbated fiscal deficits, resulting in a gross nominal public debt path that is higher than at the time of the 2019 Article IV consultation and the 2020 request for access to the RCF. However, expensive commercial loans have been replaced with official multilateral and bilateral financing at more favorable conditions which helps improve external debt burden indicators. Public and publicly guaranteed (PPG) external debt figures used for this DSA are consistent with the information in the WBG International Debt Statistics.

Text Figure 1. Papua New Guinea: Debt Owed by the Government



Source: PNG Authorities; IMF.

Source: World Bank Group International Debt Statistics; IMF.

3. The IMF's general allocation of Special Drawing Rights (SDRs) became effective in August 2021, with SDR 252 million (US$357 million, or 95.7 percent of quota) allocated to PNG. The authorities have decided to use the SDR allocation in its entirety to support the 2021 budget. The SDR allocation is included in public debt numbers for the purpose of this DSA (international reserves with the central bank increase only temporary for a few months until the SDRs are converted into hard currency and used by the treasury for budget support), and the associated debt service of 0.05 percent of the amount outstanding is reflected as well.

4. PNG is an IDA blend country, with total IDA19 allocation is SDR 178.4 million. IDA's Performance-Based Allocation (PBA) for PNG benefitted from resource frontloading amid a shortening of IDA19 to two years as well as an intra-regional reallocation for SDR 0.6 million, approved in November 2021. The IDA decision to graduate a country to IBRD-only status is based on an assessment of the country's macroeconomic prospects, risk of debt distress, vulnerability to shocks, institutional constraints, and levels of poverty and social indicators. Ongoing efforts under the Sustainable Development Finance Policy (SDFP) can help improve PNG's creditworthiness and allow future access to full IBRD borrowing. As part of the SDFP, PNG has satisfactorily implemented the Performance and Policy Actions (PPA) for fiscal year 2021.

5. PNG has participated in the G-20 Debt Service Suspension Initiative (DSSI) and the suspension granted by lenders is included in this DSA. The following amounts have been considered in the analysis: a total of US$ 312.4 million for May-December 2020, and US$ 35.8 million for January-June 2021. Prospective debt service suspension between July and December 2021 is not included.

6. The IDA Sustainable Development Finance Policy (SDFP) supports PNG in addressing key debt vulnerabilities. Fiscal risks from explicit and implicit contingent liabilities of the state-owned enterprise (SOE) sector are substantial. To improve the reporting of state guarantees, the World Bank is providing technical support to PNG's Department of Treasury and help amend the 2016 State Guarantee Policy document with a view to improve debt transparency and reporting practices.

BACKGROUND ON MACROECONOMIC FORECASTS

7. Economic activity in 2020 was significantly disrupted by the pandemic (*Text Table 2*). Real GDP in 2020 is estimated to have contracted by 3 ½ percent due to weak demand for PNG's key exports and extended mine closures as well as the effects of mobility restrictions and changes in consumption patterns. The pre-crisis forecast (2019 Article IV Staff Report) had 2020 GDP growth at 2.0 percent. Headline inflation (period-average CPI) rose to 4.9 percent in 2020, reflecting spikes in food and medical prices and the depreciation of the kina. As lower export revenues were partly offset by weaker import demand (down 19 percent), the 2020 current account surplus remained largely unchanged from 2019. Gross international reserves increased by 17 percent from 2019. However, the budget had to be supported by official bilateral and multilateral partners, including by the Fund through a disbursement under the RCF.

8. The resurgence of COVID-19 weakens the outlook for 2021. PNG has one of the lowest COVID-19 vaccination rates in the world (merely 2 percent of the population is fully vaccinated, estimated based on number of doses administered, as of mid-November 2021), owing mainly to demand factors, particularly, widespread vaccine hesitancy. In the beginning of 2021, the country was hit by two consecutive COVID-19 waves which, among other things, caused reductions in resource sector output. By July, Covid-19 case numbers dropped considerably, suggesting a recovery in resource sector production. But already in September, a third COVID-19 wave hit PNG, with record case numbers, and resource production could not resume as anticipated, consequently lowering expectations for 2021 export receipts. Given the persistence of the COVID-19 shock on PNG, domestic demand continues to be subdued, with substantially weaker import growth than forecasted earlier. At 5.0 percent, 2021 inflation (period average consumer price index)

is expected to turn out significantly higher than projected in the previous DSA (3.7 percent), driven by high medical and food prices as well as persistent bottlenecks in global supply chains. The current account surplus continues to be large, and above 20 percent, owing mainly to strong global demand for PNG's exports goods. Increased spending needs to battle the COVID-19 pandemic have driven the primary fiscal deficit to an estimated 6 percent in 2021.

Text Table 2. Papua New Guinea: Debt Sustainability Analysis Vintages: Macroeconomic and Fiscal Assumptions

	DSA Vintage	2020	2021	2022	2023	2024- 1/
Real GDP growth y/y (in percent)	**2021**	**-3.5**	**1.7**	**4.8**	**4.3**	**3.1**
	2020	-1.7	0.3	1.4	3.7	3.2
Resource sector	2021	-5.5	6.1	14.0	7.1	0.2
	2020	-13.8	14.6	9.5	4.3	3.4
Non-resource sector	2021	4.7	10.1	6.6	8.8	7.7
	2020	2.9	4.2	6.0	8.4	6.8
Memo: Pre-crisis medium-term projection 2/		2.0	2.8	2.9	3.2	
Inflation, annual average (consumer prices, percent)	**2021**	**4.9**	**5.0**	**5.6**	**4.7**	**4.5**
	2020	3.7	3.6	4.9	4.1	2.9
Current account balance (percent of GDP)	**2021**	**21.2**	**20.2**	**22.6**	**21.0**	**18.5**
	2020	14.6	18.6	17.6	16.5	14.0
Growth of exports of G&S (US$, in percent)	2021	-20.4	11.0	15.1	4.7	3.2
	2020	-10.1	15.9	3.7	3.2	2.9
Growth of imports of G&S (US$, in percent)	2021	-20.7	2.2	8.7	9.4	4.4
	2020	11.9	9.3	9.4	4.5	2.7
Primary balance (percent of GDP)	**2021**	**-6.0**	**-5.1**	**-3.7**	**-2.7**	**0.0**
	2020	-3.7	-2.5	-1.7	-1.2	0.0
Government revenues (excluding grants, percent of GDP)	2021	12.5	12.7	13.8	14.0	14.9
	2020	13.3	14.7	15.1	15.4	16.6

Source: PNG authorities; IMF staff estimates and projections.

Notes:

1/ Average refers to 2024-2041 (2021 DSA) and 2024-2040 (2020 DSA).

2/ Pre-crisis projection in 2019 Article IV Staff Report.

9. Fiscal assumptions for 2022 under the SMP include a gradual consolidation while ensuring space for needed spending is preserved (*Text Table 3*). PNG's National Economic Council (NEC) has approved a fiscal framework that targets an overall deficit of 6 percent of GDP, which has been published in the government's 2022 Budget Strategy Paper. This framework underpins the budget legislation and envisages an improvement in domestic revenues, driven by a cyclical improvement in tax revenues and policy measures taken to increase the contribution from SOEs following the passage of the Non-Tax Revenue Bill while expenditures will be focused on supporting the recovery, continuing the shift toward capital spending, adequately financing education, and the one-off allocation for the organization of the elections. Macroeconomic performance in 2022 will ultimately depend on how successfully PNG can roll out vaccines and incentivize vaccination.

Text Table 3. Papua New Guinea: Summary of Fiscal Operations, 2020–22
(Percent of GDP, unless otherwise indicated)

	2020	2021		2022	Difference [1/]
	Actual	Budget	Revised	Prog.	pp.
Revenue	**14.2**	**14.0**	**14.4**	**15.6**	**1.2**
Tax revenue	11.5	12.0	11.7	12.3	0.6
Of which: From cyclical improvements					*0.6*
Of which: From policy adjustments					*0.0*
Non-tax revenue	1.0	0.9	1.0	1.5	0.5
Of which: From cyclical improvements					*0.1*
Of which: From policy adjustments					*0.4*
Grants	1.7	1.1	1.8	1.8	0.0
Expenditure	**22.7**	**21.2**	**21.8**	**21.6**	**-0.2**
Expense	18.6	17.1	17.6	16.6	-0.9
Compensation of employees	6.8	6.2	6.4	6.0	-0.4
Use of goods and services	6.3	6.3	6.5	6.0	-0.4
Of which: COVID-19 spending [2/]	0.6	0.6	0.6	0.0	-0.6
Of which: General election expenditure				0.8	0.8
Interest	2.5	2.4	2.2	2.3	0.1
Others [3/]	2.9	2.2	2.5	2.3	-0.2
Net acquisition of non-financial assets	4.1	4.1	4.2	5.0	0.8
Net lending (+)/borrowing (-)	**-8.6**	**-7.1**	**-7.4**	**-6.0**	**1.3**
Memo item:					
Nominal GDP (billions of kina)	85.3	92.7	93.1	101.0	

Sources: Country authorities, IMF staff.

[1/] Difference for 2022 relative to the revised fiscal projection for 2021.

[2/] COVID-19 related spending was allocated as a contingency in the 2021 Budget and is being booked as spending on goods and services. Some reallocation could occur when the Final Budget Outcome is reported in 2022.

[3/] Includes grants to regional governments, social spending and other current spending not elsewhere classified.

10. Medium- to long-term projections include higher inflation and fiscal deficits—but also stronger exports growth and continued large current account surpluses. At 3.1 percent, the long-term potential real growth estimate remains essentially unchanged from the 2020 DSA. Inflation is projected to remain significantly higher, however, at around 5 percent annual average, owing largely to imported inflation from global markets. As strong global demand for PNG's export goods is expected to persist over the medium-term, and with resource-sector projects that have been delayed by the pandemic coming on stream, the current account surplus is forecast to remain very large. The macroeconomic scenario further anticipates continued primary deficits but has built in a significant amount of fiscal consolidation compared to the 6 percent deficit projected for 2021, consistent with the SMP targets. Progress in boosting revenue generation has not lived up to expectations, and the projections now assume both slower and smaller increases in revenues, including over the long-term.

11. Link between economic growth, public debt, and fiscal policy: The main near-term risk continues to be the pandemic. A new variant (Omicron) that started spreading globally adds to these concerns. In the absence of faster progress on vaccinations, further restrictions and lockdowns are possible,

with the attendant impact on growth and external inflows, particularly if the resource sector is forced to shut down again due to a high number of infections. With limited sources of financing available in an adverse scenario and continued pressing social and development needs, the room for significant policy adjustment is relatively limited. If growth deteriorated significantly compared to the projections, further debt buildup may be needed to finance the budget and maintain government services. On the upside, higher commodity prices or a decision to go ahead with several resource projects, including Papua LNG, P'nyang LNG and the Wafi Golpu mining project, could boost growth and fiscal revenues over the medium term.

12. Financing mix: For domestic financing, the DSA assumes that the composition of T-bills and T-bonds remains unchanged compared to the past five years. For the near-term, the DSA considers existing commitments and, from 2025, the DSA assumes that, on average, one-third of the financing needs are covered from official multilateral and bilateral resources, while the rest is domestic financing, with a balanced mix of bonds and T-bills. During the SMP, there is a zero limit to new non-concessional external borrowing (NCB) by the Department of Treasury and Bank of Papua New Guinea; multilateral and official bilateral financial support is exempted from this limit. NCB may be allowed only in exceptional circumstances, which may be warranted when financing is needed for a project integral to the authorities' development program for which concessional financing is not available, or when non-concessional borrowing is used for debt management operations that improve the overall public debt profile. For 2022, IDA operations of SDR 111 million are envisioned to be delivered and to be entirely financed through PNG's PBA. This DSA assumes that IDA financing continues to be extended on blend credit terms.

COUNTRY CLASSIFICATION AND DETERMINATION OF SCENARIO STRESS TEST

13. PNG's debt carrying capacity is assessed as weak. According to the October 2021 World Economic Outlook and Country Policy and Institutional Assessment (CPIA), PNG's Composite Indicator (CI) index is 2.63, indicating weak debt-carrying capacity (*Text Table 4*). Hence, the applicable thresholds are 30 percent for the present value (PV) of external debt-to-GDP ratio; 140 percent for the PV of the external debt-to-exports ratio; 10 percent for the external debt service-to-exports ratio; 14 percent for the external debt service-to-revenue ratio; and 35 percent for the PV of public debt-to GDP ratio, respectively.

Text Table 4. Papua New Guinea: Composite Indicator and Applicable Thresholds

Calculation of the CI Index

Components	Coefficients (A)	10-year average values (B)	CI Score components (A*B) = (C)	Contribution of components
CPIA	0.385	2.966	1.14	43%
Real growth rate (in percent)	2.719	2.629	0.07	3%
Import coverage of reserves (in percent)	4.052	42.419	1.72	65%
Import coverage of reserves^2 (in percent)	-3.990	17.994	-0.72	-27%
Remittances (in percent)	2.022	0.366	0.01	0%
World economic growth (in percent)	13.520	3.078	0.42	16%
CI Score			2.64	100%
CI rating			Weak	

Applicable thresholds

APPLICABLE	
EXTERNAL debt burden thresholds	
PV of debt in % of	
Exports	140
GDP	30
Debt service in % of	
Exports	10
Revenue	14

APPLICABLE	
TOTAL public debt benchmark	
PV of total public debt in percent of GDP	35

14. Scenario stress tests: As indicated in the section on public debt coverage, a contingent liabilities stress test is included to account for SOE debt not captured in official public debt data. In the context of the previous SMP, which expired in end-June 2021, the authorities initiated an SOE reform program to reduce the backlog of audited annual financial statements and to strengthen SOE oversight and improve understanding of fiscal risks. This program includes a detailed review of SOE debt and government guarantees and is also expected to improve the reporting of public debt. Further, given the size and importance of PNG's resource sector (with a share of commodities in total exports of goods and services of 96 percent), a commodity price shock is included in the DSA. Considering the high price volatility over the past two year, the fuel price shock is set at 35 percent (compared to the default shock of 27 percent) and the shock to non-fuel commodity prices is set to 21 percent—with 20 percent for base metals and precious metals, and 22 percent for agricultural commodities other than grain (price shocks to grain like wheat, corn, and soybeans, are not relevant for PNG and, therefore, not included in the stress test). Mitigating factors are included as well, and at default values (2 percent for fuel, and 27 percent for non-fuel). Separately, a third stress test is included to simulate market financing pressures, using standard assumptions, including for exchange rate depreciation (15 percent).

DEBT SUSTAINABILITY

A. External Debt Sustainability Analysis

15. Under the baseline scenario, the debt-service to revenue indicator is projected to breach its threshold. There is a small breach in 2026, and a relatively large breach in 2028, when a bullet payment for the 2018 Eurobond is due. The authorities do not plan, at least not at this juncture, to roll over this bond. After 2028, the indicator remains close to the threshold and the marginal breaches of the threshold are merely short-lived (one year). The projections further show one small and short-lived breach of the debt service-to-exports ratio, which is discounted as well, and on the same grounds. The present value of the debt-to-GDP ratio as well as the debt-to-exports ratio are below their respective thresholds over the entire projection horizon, and on a clear downward trend.

16. Stress tests point to vulnerabilities in PNG's external debt dynamics particularly with respect to exports shocks, which would cause threshold breaches for all four external sustainability indicators. Changes in policy and the structure of the economy manifests in a divergence between the historical scenario and the baseline.

17. As in the past, the assessment of debt dynamics is hampered by large residuals from external financial flows from money transfer by resource companies via offshore accounts.

B. Public Sector Debt Sustainability Analysis

18. Public debt ratios have increased substantially in recent years and reached almost 50 percent of GDP in 2020. Starting from this level means that the public debt sustainability indicator is in breach of the threshold for countries with weak debt-carrying capacity (that is, 35 percent of GDP) during the first half of the projection horizon but, with public debt coming down, meets the threshold from 2031. Similarly, the debt-to-revenue ratio sees a continuous downward trend over the projection horizon.

RISK RATINGS AND VULNERABILITIES

19. PNG remains at "high" risk of external and overall debt distress. The (mechanical) external debt distress rating as well as the overall debt distress rating are "high", owing to the breaches of sustainability thresholds discussed in the previous section. No staff judgement has been applied to these ratings.

20. Debt service on existing loans, paired with relatively weak revenue generation, are expected to almost double the debt service-to-revenue ratio over the medium term. However, as debt service reduces and revenues increase, and barring further shocks to demand growth, the indicator enters a significant downward trend from its peak in 2026.

21. Stress tests show that adverse shocks to exports constitute the main risk to public debt sustainability. Further, the historical scenario indicates that it will be challenging to reduce debt from

current levels and that reforms, including those envisaged in the SMP, are essential for supporting the sustainability of public finances.

22. Debt dynamics are assessed as sustainable. Public debt is expected not to increase significantly in the near term and to enter a clear downward path over the medium term. Also, the projected temporary breaches of sustainability indicators can be prevented by debt management operations as well as by boosting revenue generation. The external debt-to-GDP and debt-to-exports ratios are below their thresholds over the entire projection horizon. Public external and overall debt is judged to be sustainable conditional on the implementation of the policy measures part of the SMP.

AUTHORITIES' VIEWS

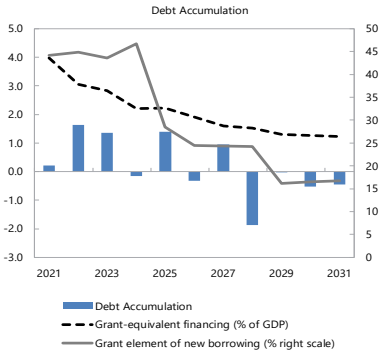
23. The authorities took note of Staff's assessment that PNG remains at high risk of debt distress. In discussions, the authorities agree that large financing requirements to dampen the effects of the COVID-19 pandemic on PNG's economy have left their mark on public debt. Fiscal consolidation and conservative financing strategies will help support the sustainability of PNG's debt going forward.

Table 1. Papua New Guinea: External Debt Sustainability Framework, Baseline Scenario, 2018-2041

(In percent of GDP, unless otherwise indicated)

	Actual			Projections								Average 8/	
	2018	2019	2020	2021	2022	2023	2024	2025	2026	2031	2041	Historical	Projections
External debt (nominal) 1/	**71.4**	**62.6**	**69.2**	**68.1**	**65.9**	**63.8**	**60.9**	**59.7**	**57.8**	**45.0**	**25.9**	**75.7**	**57.2**
of which: public and publicly guaranteed (PPG)	**15.2**	**18.0**	**23.0**	**24.9**	**25.3**	**25.5**	**24.2**	**24.4**	**23.9**	**15.9**	**5.5**	**6.7**	**22.4**
Change in external debt	-3.7	-8.9	6.6	-1.1	-2.2	-2.1	-2.9	-1.2	-1.9	-2.7	-2.0		
Identified net debt-creating flows	**-28.6**	**-24.7**	**-20.2**	**-21.3**	**-25.6**	**-23.6**	**-22.2**	**-21.4**	**-20.5**	**-17.3**	**-11.2**	**-10.0**	**-20.8**
Non-interest current account deficit	**-27.0**	**-25.5**	**-23.0**	**-24.0**	**-24.2**	**-22.5**	**-21.6**	**-20.7**	**-20.8**	**-17.3**	**-11.2**	**-9.2**	**-20.9**
Deficit in balance of goods and services	-25.1	-26.5	-21.5	-23.3	-25.6	-24.1	-23.0	-22.2	-21.1	-17.9	-11.6	**-9.3**	**-21.5**
Exports	41.7	47.6	38.4	39.1	41.5	40.2	38.7	37.5	35.4	30.6	22.9		
Imports	16.6	21.0	16.9	15.9	15.9	16.0	15.7	15.3	14.3	12.7	11.3		
Net current transfers (negative = inflow)	-1.5	-1.3	-1.4	-1.0	-1.0	-0.9	-1.2	-1.1	-1.1	-0.8	-0.4	**-1.0**	**-1.0**
of which: official	-0.5	-1.4	-1.2	-0.9	-0.9	-0.8	-0.8	-0.7	-0.7	-0.5	-0.3		
Other current account flows (negative = net inflow)	-0.4	2.3	0.0	0.3	2.4	2.5	2.6	2.6	1.4	1.3	0.8	**1.2**	**1.6**
Net FDI (negative = inflow)	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**
Endogenous debt dynamics 2/	**-1.6**	**0.8**	**2.8**	**2.7**	**-1.4**	**-1.1**	**-0.6**	**-0.8**	**0.3**	**0.0**	**0.1**		
Contribution from nominal interest rate	2.7	2.6	1.8	3.8	1.6	1.5	1.2	1.0	2.0	1.4	0.9		
Contribution from real GDP growth	0.2	-3.1	2.2	-1.1	-3.0	-2.6	-1.8	-1.7	-1.7	-1.4	-0.9		
Contribution from price and exchange rate changes	-4.5	1.3	-1.2	...	...	...	...	...	...	...	...		
Residual 3/	**25.0**	**15.9**	**26.8**	**20.3**	**23.4**	**21.5**	**19.3**	**20.2**	**18.6**	**14.6**	**9.1**	**12.8**	**18.6**
of which: exceptional financing	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Sustainability indicators													
PV of PPG external debt-to-GDP ratio	**...**	**...**	**28.4**	**26.2**	**25.7**	**25.0**	**23.3**	**23.3**	**21.7**	**14.3**	**5.7**		
PV of PPG external debt-to-exports ratio	**...**	**...**	**73.9**	**67.0**	**61.9**	**62.2**	**60.4**	**62.2**	**61.2**	**46.8**	**24.9**		
PPG debt service-to-exports ratio	**1.6**	**2.2**	**8.5**	**10.0**	**3.0**	**3.9**	**5.4**	**4.3**	**8.1**	**7.0**	**5.4**		
PPG debt service-to-revenue ratio	**4.2**	**7.3**	**26.1**	**30.9**	**9.0**	**11.1**	**14.6**	**11.0**	**19.8**	**14.0**	**8.6**		
Gross external financing need (Million of U.S. dollars)	-3883.1	-3472.8	-3187.8	-3515.6	-4815.4	-4533.6	-4760.4	-4938.7	-5762.5	-5700.6	-7396.7		
Key macroeconomic assumptions													
Real GDP growth (in percent)	-0.3	4.5	-3.5	1.7	4.8	4.3	3.0	3.0	3.0	3.0	3.3	**3.9**	**3.2**
GDP deflator in US dollar terms (change in percent)	6.3	-1.7	2.0	7.2	3.6	3.8	3.2	3.1	3.1	3.3	3.4	**2.1**	**3.6**
Effective interest rate (percent) 4/	3.8	3.8	2.8	5.9	2.6	2.5	2.0	1.7	3.6	3.1	3.5	**3.0**	**3.3**
Growth of exports of G&S (US dollar terms, in percent)	-2.5	17.2	-20.4	11.0	15.1	4.7	2.3	2.9	0.3	2.6	8.7	**6.0**	**4.8**
Growth of imports of G&S (US dollar terms, in percent)	6.4	30.3	-20.7	2.2	8.7	9.4	3.7	3.7	-0.7	3.9	15.7	**-1.3**	**4.2**
Grant element of new public sector borrowing (in percent)	...	...	...	44.1	44.8	43.6	46.7	28.5	24.5	16.8	13.8	**...**	**30.0**
Government revenues (excluding grants, in percent of GDP)	15.4	14.2	12.5	12.7	13.8	14.0	14.4	14.7	14.5	15.2	14.3	**14.4**	**14.3**
Aid flows (in Million of US dollars) 5/	1431.2	1258.7	1731.2	1347.8	1084.8	1116.4	754.8	749.1	693.2	521.5	693.4		
Grant-equivalent financing (in percent of GDP) 6/	...	...	...	4.0	3.1	2.8	2.2	2.2	1.9	1.2	0.8	**...**	**2.1**
Grant-equivalent financing (in percent of external financing) 6/	...	...	...	58.7	66.5	63.7	74.0	54.4	56.3	55.9	56.7	**...**	**59.0**
Nominal GDP (Million of US dollars)	24,110	24,751	24,364	26,573	28,836	31,235	33,182	35,223	37,403	50,665	96,038		
Nominal dollar GDP growth	6.0	2.7	-1.6	9.1	8.5	8.3	6.2	6.2	6.2	6.4	6.8	**5.9**	**6.9**
Memorandum items:													
PV of external debt 7/	...	...	74.6	69.4	66.3	63.2	60.1	58.6	55.5	43.4	26.1		
In percent of exports	...	...	194.1	177.4	159.6	157.4	155.4	156.4	157.0	141.9	114.1		
Total external debt service-to-exports ratio	26.3	24.1	25.8	27.5	18.1	19.9	18.8	17.7	15.2	19.9	15.4		
PV of PPG external debt (in Million of US dollars)			6923.9	6974.2	7408.0	7798.3	7747.7	8211.5	8100.8	7258.1	5484.8		
(PVt-PVt-1)/GDPt-1 (in percent)				0.2	1.6	1.4	-0.2	1.4	-0.3	-0.5	-0.3		
Non-interest current account deficit that stabilizes debt ratio	-23.4	-16.6	-29.6	-22.9	-22.0	-20.4	-18.7	-19.5	-18.9	-14.6	-9.2		

Definition of external/domestic debt	Residency-based
Is there a material difference between the two criteria?	No



Sources: Country authorities; and staff estimates and projections.

1/ Includes both public and private sector external debt.

2/ Derived as [r - g - ρ(1+g) + εα (1+r)]/(1+g+ρ+gρ) times previous period debt ratio, with r = nominal interest rate; g = real GDP growth rate, ρ = growth rate of GDP deflator in U.S. dollar terms, ε=nominal appreciation of the local currency, and α= share of local currency-denominated external debt in total external debt.

3/ Includes exceptional financing (i.e., changes in arrears and debt relief); changes in gross foreign assets; and valuation adjustments. For projections also includes contribution from price and exchange rate changes.

4/ Current-year interest payments divided by previous period debt stock.

5/ Defined as grants, concessional loans, and debt relief.

6/ Grant-equivalent financing includes grants provided directly to the government and through new borrowing (difference between the face value and the PV of new debt).

7/ Assumes that PV of private sector debt is equivalent to its face value.

8/ Historical averages are generally derived over the past 10 years, subject to data availability, whereas projections averages are over the first year of projection and the next 10 years.

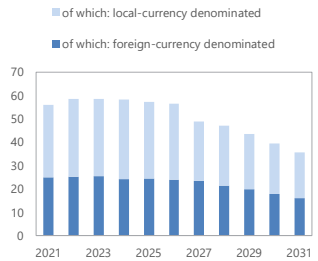
Table 2. Papua New Guinea: Public Debt Sector Sustainability Framework, Baseline Scenario, 2018-2041

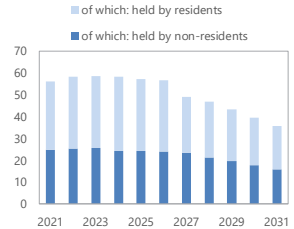
(In percent of GDP, unless otherwise indicated)

	Actual			Projections								Average 6/	
	2018	2019	2020	2021	2022	2023	2024	2025	2026	2031	2041	Historical	Projections
Public sector debt 1/	**36.7**	**41.5**	**49.0**	**56.0**	**58.4**	**58.5**	**58.3**	**57.1**	**56.6**	**35.6**	**14.4**	**26.3**	**50.8**
of which: external debt	**15.2**	**18.0**	**23.0**	**24.9**	**25.3**	**25.5**	**24.2**	**24.4**	**23.9**	**15.9**	**5.5**	**6.7**	**22.4**
Change in public sector debt	2.0	4.7	7.5	7.0	2.4	0.1	-0.2	-1.2	-0.5	-3.8	-0.7		
Identified debt-creating flows	**0.0**	**2.8**	**8.4**	**6.9**	**2.8**	**0.6**	**0.1**	**-0.8**	**-0.1**	**-3.3**	**-1.5**	**3.8**	**-0.8**
Primary deficit	**0.2**	**1.9**	**6.0**	**5.1**	**3.7**	**2.7**	**1.1**	**0.0**	**-0.6**	**-2.6**	**-1.2**	**2.5**	**0.1**
Revenue and grants	17.7	16.3	14.2	14.4	15.6	15.6	15.9	16.2	15.9	16.2	15.0	**16.1**	**15.6**
of which: grants	2.3	2.1	1.7	1.8	1.8	1.6	1.5	1.5	1.4	1.0	0.7		
Primary (noninterest) expenditure	18.0	18.2	20.2	19.6	19.3	18.3	17.1	16.3	15.3	13.6	13.8	**18.6**	**15.8**
Automatic debt dynamics	**-0.2**	**0.9**	**2.4**	**1.8**	**-1.0**	**-2.1**	**-1.0**	**-0.9**	**0.5**	**-0.7**	**-0.3**		
Contribution from interest rate/growth differential	0.1	0.7	2.6	1.8	-1.0	-2.1	-1.0	-0.9	0.5	-0.7	-0.3		
of which: contribution from average real interest rate	0.0	2.2	1.1	2.6	1.6	0.4	0.7	0.8	2.1	0.4	0.2		
of which: contribution from real GDP growth	0.1	-1.6	1.5	-0.8	-2.6	-2.4	-1.7	-1.7	-1.7	-1.2	-0.5		
Contribution from real exchange rate depreciation	-0.3	0.3	-0.2	...	...	...	...	...	...	...	...		
Other identified debt-creating flows	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**
Privatization receipts (negative)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Recognition of contingent liabilities (e.g., bank recapitalization)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Debt relief (HIPC and other)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Other debt creating or reducing flow (please specify)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Residual	**2.0**	**1.9**	**-0.9**	**0.1**	**-0.4**	**-0.5**	**-0.3**	**-0.4**	**-0.4**	**-0.5**	**0.8**	**1.0**	**-0.4**
Sustainability indicators													
PV of public debt-to-GDP ratio 2/	**...**	**...**	**54.5**	**57.4**	**58.8**	**58.0**	**57.5**	**56.1**	**54.3**	**34.1**	**14.8**		
PV of public debt-to-revenue and grants ratio	**...**	**...**	**384.6**	**397.8**	**377.5**	**371.7**	**361.6**	**345.5**	**341.3**	**210.4**	**99.0**		
Debt service-to-revenue and grants ratio 3/	**103.0**	**109.8**	**140.0**	**65.9**	**75.2**	**102.2**	**114.0**	**124.8**	**134.9**	**91.7**	**41.7**		
Gross financing need 4/	18.5	19.8	25.9	14.6	15.4	18.6	19.3	20.3	20.9	12.2	5.1		
Key macroeconomic and fiscal assumptions													
Real GDP growth (in percent)	-0.3	4.5	-3.5	1.7	4.8	4.3	3.0	3.0	3.0	3.0	3.3	**3.9**	**3.2**
Average nominal interest rate on external debt (in percent)	2.6	3.9	2.7	13.0	3.4	3.7	3.2	2.9	8.1	1.3	1.9	**3.0**	**4.4**
Average real interest rate on domestic debt (in percent)	-0.2	9.4	3.5	0.4	4.1	-0.3	1.1	1.8	2.2	2.3	1.9	**4.6**	**0.5**
Real exchange rate depreciation (in percent, + indicates depreciation)	-2.8	1.9	-1.2	...	...	...	...	...	...	...	...	**1.0**	**...**
Inflation rate (GDP deflator, in percent)	9.8	1.1	5.5	7.2	3.6	3.8	3.2	3.1	3.1	3.3	3.4	**4.3**	**5.4**
Growth of real primary spending (deflated by GDP deflator, in percent)	10.3	5.8	7.0	-1.5	3.5	-1.4	-3.8	-1.9	-2.7	3.1	3.5	**4.1**	**-0.4**
Primary deficit that stabilizes the debt-to-GDP ratio 5/	-1.7	-2.9	-1.5	-1.8	1.4	2.5	1.3	1.2	0.0	1.2	-0.5	**-2.0**	**1.4**
PV of contingent liabilities (not included in public sector debt)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		

Definition of external/domestic debt	**Residency-based**
Is there a material difference between the two criteria?	**No**

Public sector debt 1/





Sources: Country authorities; and staff estimates and projections.

1/ Coverage of debt: The central, state, and local governments, government-guaranteed debt . Definition of external debt is Residency-based.

2/ The underlying PV of external debt-to-GDP ratio under the public DSA differs from the external DSA with the size of differences depending on exchange rates projections.

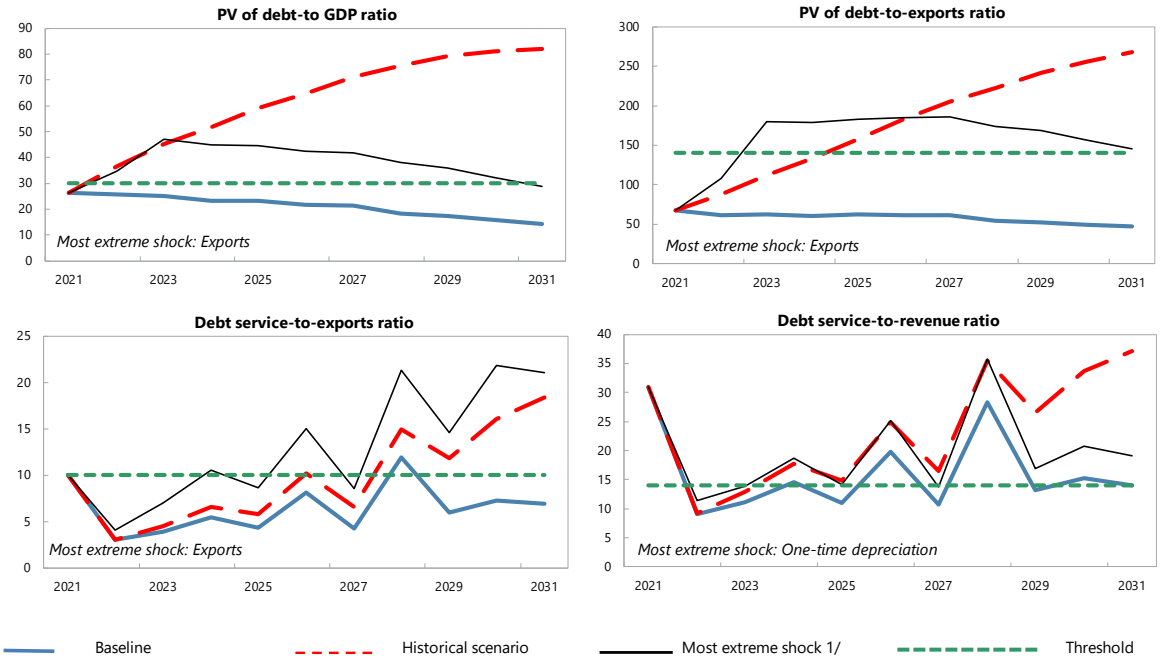
3/ Debt service is defined as the sum of interest and amortization of medium and long-term, and short-term debt.

4/ Gross financing need is defined as the primary deficit plus debt service plus the stock of short-term debt at the end of the last period and other debt creating/reducing flows.

5/ Defined as a primary deficit minus a change in the public debt-to-GDP ratio ((-): a primary surplus), which would stabilizes the debt ratio only in the year in question.

6/ Historical averages are generally derived over the past 10 years, subject to data availability, whereas projections averages are over the first year of projection and the next 10 years.

Figure 1. Papua New Guinea: Indicators of Public and Publicly Guaranteed External Debt under Alternative Scenarios, 2021-2031



Customization of Default Settings	Size	Interactions
Tailored Stress		
Combined CL	Yes	
Natural disaster	n.a.	n.a.
Commodity price	Yes	No
Market financing	No	No

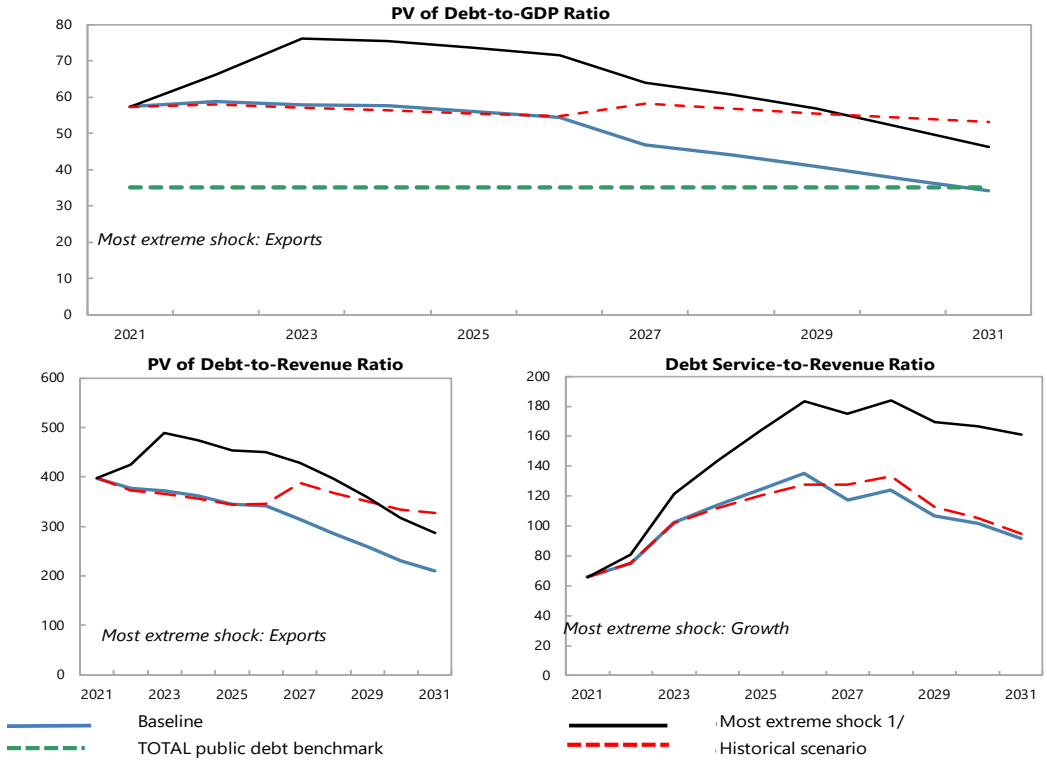
Note: "Yes" indicates any change to the size or interactions of the default settings for the stress tests. "n.a." indicates that the stress test does not apply.

Borrowing assumptions on additional financing needs resulting from the stress tests*	Default	User defined
Shares of marginal debt		
External PPG MLT debt	100%	
Terms of marginal debt		
Avg. nominal interest rate on new borrowing in USD	1.3%	1.3%
USD Discount rate	5.0%	5.0%
Avg. maturity (incl. grace period)	17	17
Avg. grace period	6	6

* Note: All the additional financing needs generated by the shocks under the stress tests are assumed to be covered by PPG external MLT debt in the external DSA. Default terms of marginal debt are based on baseline 10-year projections.

Sources: Country authorities; and staff estimates and projections.
1/ The most extreme stress test is the test that yields the highest ratio in or before 2031. The stress test with a one-off breach is also presented (if any), while the one-off breach is deemed away for mechanical signals. When a stress test with a one-off breach happens to be the most exterme shock even after disregarding the one-off breach, only that stress test (with a one-off breach) would be presented.

Figure 2. Papua New Guinea: Indicators of Public Debt Under Alternative Scenarios, 2021-2031



Borrowing assumptions on additional financing needs resulting from the stress tests*	Default	User defined
Shares of marginal debt		
External PPG medium and long-term	13%	13%
Domestic medium and long-term	42%	42%
Domestic short-term	45%	45%
Terms of marginal debt		
External MLT debt		
Avg. nominal interest rate on new borrowing in USD	1.3%	1.3%
Avg. maturity (incl. grace period)	17	17
Avg. grace period	6	6
Domestic MLT debt		
Avg. real interest rate on new borrowing	2.0%	2.0%
Avg. maturity (incl. grace period)	2	2
Avg. grace period	1	1
Domestic short-term debt		
Avg. real interest rate	0.2%	0.2%

* Note: The public DSA allows for domestic financing to cover the additional financing needs generated by the shocks under the stress tests in the public DSA. Default terms of marginal debt are based on baseline 10-year projections.

Sources: Country authorities; and staff estimates and projections.

1/ The most extreme stress test is the test that yields the highest ratio in or before 2031. The stress test with a one-off breach is also presented (if any), while the one-off breach is deemed away for mechanical signals. When a stress test with a one-off breach happens to be the most exterme shock even after disregarding the one-off breach, only that stress test (with a one-off breach) would be presented.

Table 3. Papua New Guinea: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2021-2031

	Projections 1/										
	2021	2022	2023	2024	2025	2026	2027	2028	2029	2030	2031
PV of debt-to GDP ratio											
Baseline	26	26	25	23	23	22	21	18	17	16	14
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2021-2031 2/	26	**36**	**45**	**52**	**59**	**65**	**71**	**75**	**79**	**81**	**82**
B. Bound Tests											
B1. Real GDP growth	26	28	29	27	27	25	25	21	20	18	17
B2. Primary balance	26	26	26	24	24	23	23	20	19	17	16
B3. Exports	26	**35**	**47**	**45**	**44**	**42**	**42**	**38**	**36**	**32**	29
B4. Other flows 3/	26	26	26	24	24	22	22	19	18	16	15
B5. Depreciation	26	**32**	**34**	**32**	**32**	29	29	25	24	22	20
B6. Combination of B1-B5	26	**34**	**36**	**34**	**34**	**31**	**31**	27	26	23	21
C. Tailored Tests											
C1. Combined contingent liabilities	26	27	27	26	27	26	26	23	23	21	20
C2. Natural disaster	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C3. Commodity price	26	**31**	**35**	**34**	**35**	**34**	**34**	**32**	**30**	29	27
C4. Market Financing	26										
Threshold	30	30	30	30	30	30	30	30	30	30	30
PV of debt-to-exports ratio											
Baseline	67	62	62	60	62	61	62	54	52	49	47
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2021-2031 2/	67	88	112	134	**157**	**183**	**206**	**223**	**241**	**256**	**268**
B. Bound Tests											
B1. Real GDP growth	67	62	62	60	62	61	62	54	52	49	47
B2. Primary balance	67	62	64	63	65	65	66	58	57	55	52
B3. Exports	67	108	**180**	**179**	**183**	**184**	**185**	**173**	**169**	**156**	**145**
B4. Other flows 3/	67	63	64	62	64	63	63	56	54	51	48
B5. Depreciation	67	62	67	65	67	66	67	59	58	54	51
B6. Combination of B1-B5	67	89	78	101	104	103	104	94	90	84	79
C. Tailored Tests											
C1. Combined contingent liabilities	67	65	67	68	72	73	75	69	69	67	66
C2. Natural disaster	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C3. Commodity price	67	89	100	99	102	102	102	96	96	93	90
C4. Market Financing	67										
Threshold	140	140	140	140	140	140	140	140	140	140	140
Debt service-to-exports ratio											
Baseline	**10**	3	4	5	4	8	4	**12**	6	7	7
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2021-2031 2/	**10**	3	5	7	6	**10**	7	**15**	**12**	**16**	**18**
B. Bound Tests											
B1. Real GDP growth	**10**	3	4	5	4	8	4	**12**	6	7	7
B2. Primary balance	**10**	3	4	6	4	8	4	**12**	6	8	7
B3. Exports	**10**	4	7	**11**	9	**15**	9	**21**	**15**	**22**	**21**
B4. Other flows 3/	**10**	3	4	5	4	8	4	**12**	6	7	7
B5. Depreciation	**10**	3	4	6	4	8	4	**12**	6	8	8
B6. Combination of B1-B5	**10**	4	5	8	6	**11**	6	**16**	10	**12**	**12**
C. Tailored Tests											
C1. Combined contingent liabilities	**10**	3	4	6	4	8	4	**12**	6	8	7
C2. Natural disaster	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C3. Commodity price	**10**	4	5	7	5	10	5	**14**	8	**11**	**11**
C4. Market Financing											
Threshold	10	10	10	10	10	10	10	10	10	10	10
Debt service-to-revenue ratio											
Baseline	**31**	9	11	**15**	11	**20**	11	**28**	13	**15**	**14**
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2021-2031 2/	**31**	9	13	**18**	**15**	**25**	**16**	**35**	**26**	**34**	**37**
B. Bound Tests											
B1. Real GDP growth	**31**	10	13	**17**	13	**23**	13	**33**	**16**	**18**	**16**
B2. Primary balance	**31**	9	11	**15**	11	**20**	11	**29**	14	**16**	**15**
B3. Exports	**31**	9	13	**18**	**14**	**24**	14	**33**	**21**	**30**	**28**
B4. Other flows 3/	**31**	9	11	**15**	11	**20**	11	**28**	13	**16**	**14**
B5. Depreciation	**31**	11	14	**19**	**14**	**25**	14	**36**	**17**	**21**	**19**
B6. Combination of B1-B5	**31**	10	13	**18**	13	**23**	13	**33**	**19**	**22**	**20**
C. Tailored Tests											
C1. Combined contingent liabilities	**31**	9	11	**15**	11	**20**	11	**29**	14	**16**	**15**
C2. Natural disaster	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C3. Commodity price	**31**	10	14	**18**	14	**23**	13	**31**	**18**	**23**	**22**
Threshold	14	14	14	14	14	14	14	14	14	14	14

Sources: Country authorities; and staff estimates and projections.
1/ A bold value indicates a breach of the threshold.
2/ Variables include real GDP growth, GDP deflator (in U.S. dollar terms), non-interest current account in percent of GDP, and non-debt creating flows.
3/ Includes official and private transfers and FDI.

Table 4. Papua New Guinea: Sensitivity Analysis for Key Indicators of Public Debt, 2021-2031

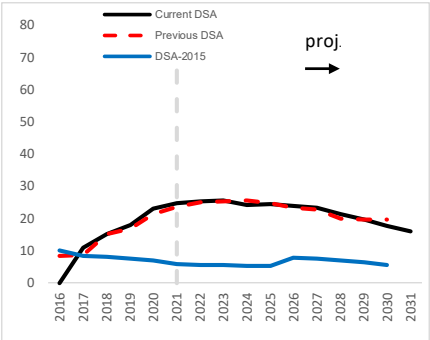
	Projections 1/										
	2021	2022	2023	2024	2025	2026	2027	2028	2029	2030	2031
PV of Debt-to-GDP Ratio											
Baseline	**57**	**59**	**58**	**58**	**56**	**54**	**47**	**44**	**41**	**37**	34
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2021-2031 2/	**57**	**58**	**57**	**56**	**56**	**55**	**58**	**57**	**56**	**54**	**53**
B. Bound Tests											
B1. Real GDP growth	**57**	**65**	**72**	**73**	**74**	**74**	**67**	**66**	**64**	**62**	**60**
B2. Primary balance	**57**	**61**	**63**	**62**	**60**	**59**	**51**	**48**	**45**	**41**	**38**
B3. Exports	**57**	**66**	**76**	**75**	**74**	**72**	**64**	**61**	**57**	**51**	**46**
B4. Other flows 3/	**57**	**59**	**59**	**58**	**57**	**55**	**47**	**44**	**41**	**38**	34
B5. Depreciation	**57**	**63**	**61**	**59**	**56**	**54**	**46**	**42**	**39**	34	30
B6. Combination of B1-B5	**57**	**59**	**61**	**61**	**60**	**58**	**51**	**48**	**46**	**42**	**39**
C. Tailored Tests											
C1. Combined contingent liabilities	**57**	**75**	**73**	**72**	**70**	**68**	**60**	**57**	**53**	**50**	**46**
C2. Natural disaster	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C3. Commodity price	**57**	**62**	**65**	**69**	**72**	**73**	**68**	**67**	**65**	**63**	**61**
C4. Market Financing	**57**										
TOTAL public debt benchmark	35	35	35	35	35	35	35	35	35	35	35
PV of Debt-to-Revenue Ratio											
Baseline	398	378	372	362	345	341	314	287	259	231	210
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2021-2031 2/	398	372	366	355	343	346	387	368	351	334	327
B. Bound Tests											
B1. Real GDP growth	398	414	451	454	448	458	444	422	401	377	365
B2. Primary balance	398	392	402	390	372	368	341	312	283	253	232
B3. Exports	398	425	488	474	454	450	428	396	360	317	286
B4. Other flows 3/	398	379	375	365	349	345	318	290	262	234	213
B5. Depreciation	398	409	393	374	351	339	312	278	246	213	189
B6. Combination of B1-B5	398	379	387	380	366	365	341	315	289	261	242
C. Tailored Tests											
C1. Combined contingent liabilities	398	482	471	455	433	428	403	370	338	305	282
C2. Natural disaster	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C3. Commodity price	398	440	461	482	476	483	465	431	410	387	375
C4. Market Financing	398										
Debt Service-to-Revenue Ratio											
Baseline	66	75	102	114	125	135	117	124	107	101	92
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2021-2031 2/	66	75	102	112	120	128	128	133	112	105	94
B. Bound Tests											
B1. Real GDP growth	66	81	122	144	164	183	175	184	170	167	161
B2. Primary balance	66	75	109	130	141	149	132	136	118	111	101
B3. Exports	66	75	103	116	127	137	119	126	112	113	102
B4. Other flows 3/	66	75	102	114	125	135	117	124	107	102	92
B5. Depreciation	66	71	99	111	119	132	111	125	104	100	89
B6. Combination of B1-B5	66	74	105	118	130	142	127	134	118	113	104
C. Tailored Tests											
C1. Combined contingent liabilities	66	75	152	178	173	183	164	162	142	132	119
C2. Natural disaster	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C3. Commodity price	66	85	117	132	154	177	169	174	162	161	157
C4. Market Financing											

Sources: Country authorities; and staff estimates and projections.
1/ A bold value indicates a breach of the benchmark.
2/ Variables include real GDP growth, GDP deflator and primary deficit in percent of GDP.
3/ Includes official and private transfers and FDI.

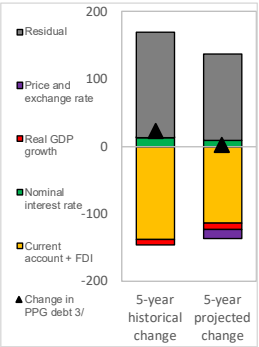
Figure 3. Papua New Guinea: Drivers of Debt Dynamics – Baseline Scenario

External debt

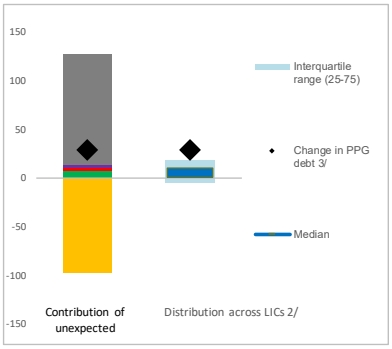
Gross Nominal PPG External Debt
(in percent of GDP; DSA vintages)



Debt-creating flows
(percent of GDP)

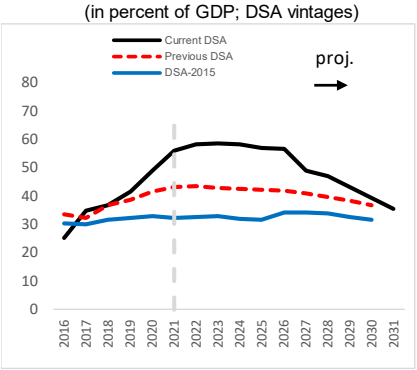


Unexpected Changes in Debt 1/
(past 5 years, percent of GDP)

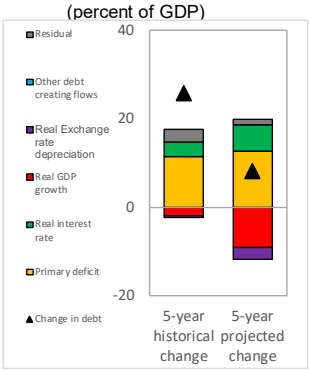


Public debt

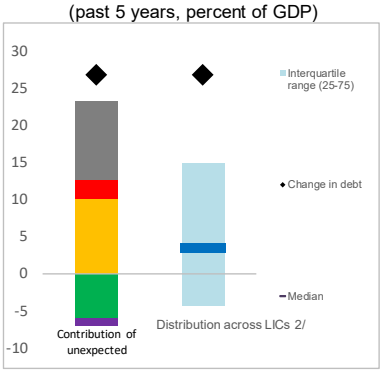
Gross Nominal Public Debt
(in percent of GDP; DSA vintages)



Debt-creating flows
(percent of GDP)



Unexpected Changes in Debt 1/
(past 5 years, percent of GDP)



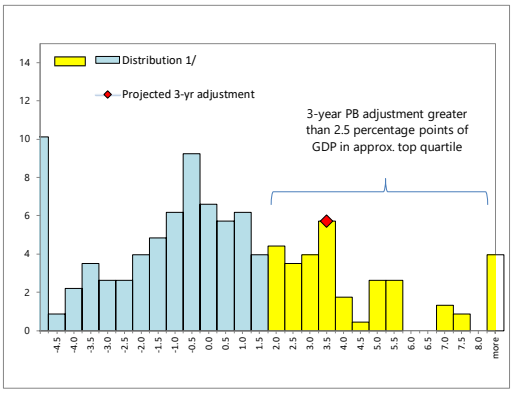
1/ Difference between anticipated and actual contributions on debt ratios.

2/ Distribution across LICs for which LIC DSAs were produced.

3/ Given the relatively low private external debt for average low-income countries, a ppt change in PPG external debt should be largely explained by the drivers of the external debt dynamics equation.

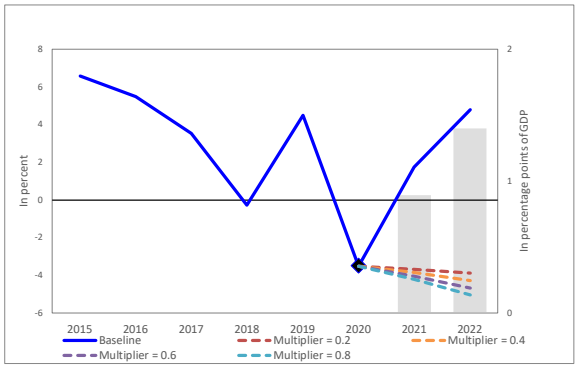
Figure 4. Papua New Guinea: Realism Tools

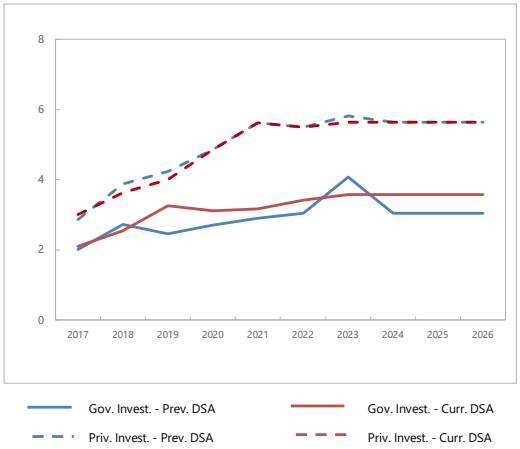
3-Year Adjustment in Primary Balance
(Percentage points of GDP)



1/ Data cover Fund-supported programs for LICs (excluding emergency financing) approved since 1990. The size of 3-year adjustment from program inception is found on the horizontal axis; the percent of sample is found on the vertical axis.

Fiscal Adjustment and Possible Growth Paths 1/



1/ Bars refer to annual projected fiscal adjustment (right-hand side scale) and lines show possible real GDP growth paths under different fiscal multipliers (left-hand side scale).

Public and Private Investment Rates
(percent of GDP)